

Ticketing System Project

(Simulating a real world IT help desk environment)

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Environment: Windows Server(DC1)|Windows OS 10 Client|
osTicket Help Desk

Purpose

The purpose of this project is to build a help desk ticketing system that simulates a real-world office environment where employees can raise tickets related to IT issues, and support agents can retrieve, troubleshoot, document, and resolve those tickets before closing them.

Through this project, I aim to demonstrate key technical and problem-solving skills required in an IT support role, such as setting up a ticketing system from scratch, configuring its core components, and understanding the complete lifecycle of a ticket from submission to resolution.

By the end of the project, I will have gained hands-on experience in how a real-world IT help desk operates, learning how technical issues are tracked, prioritized, and resolved efficiently in an organised system.

Objectives

- Download and install osTicket along with all required components, (e.g., MySQL, PHP and Apache) on the Windows Server.
- Configure osTicket to ensure it is fully functional and accessible from the browser.
- Create User Accounts and assign them to appropriate groups or departments.
- Generate one ticket to simulate real IT issues and demonstrate the troubleshooting process from creation to resolution.
- Document each ticket by clearly stating the issue, cause and solution.
- Record the skills learned and identify areas for improvement in future projects.
- Integrate the ticketing system with the Active Directory Domain Service (AD DS) environment to simulate a real enterprise setup.

Tools & Technology used

- **Virtual Box** - Application used to host all virtual Machines.
- **Windows Server 2022** - The main server where Active Directory and osTicket is hosted.
- **Windows 10 OS Client** - Workstations that users log into and use to submit tickets.
- **Active Directory Domain Services (AD DS)** - Used to manage users, groups and group policies across the domain.
- **osTicket** - Web-based ticketing system used for reporting, tracking and resolving IT issues.
- **Apache** - Web server that hosts the osTicket Application.
- **MySQL** - Database engine used to store ticket data and user information.
- **phpMyAdmin** - Graphical interface for managing the MySQL database.
- **PHP** - Server-side scripting language used by Apache to process and display osTicket content on the browser.

Project Overview

In this Project, I'm integrating my Active Directory Domain Services (AD DS) project with a ticketing system (osTicket) to simulate a real-world office environment where users can submit IT support tickets and agents can resolve them.

To ensure the project functions properly, I will download and configure the required components such as osTicket, Apache, PHP and MySQL. To simplify the installation process, I will use XAMPP, which includes all necessary components (Apache, PHP, and MySQL) in one package.

The ticketing system will be installed and configured on the Windows Server 2022, where it will run centrally. The Windows 10 client machines will access it using the server domain link (e.g., <http://10.0.2.10/osticket>).

Since osTicket is not connected to Active Directory, agents cannot log in using their domain accounts. Instead, new agents will be created within osTicket. I will create three main accounts, Admin, Agent, and User, to simulate a complete help desk workflow,

The lifecycle of the ticket in this system is as follows: a user submits a ticket, the agent investigates and resolves the issue, and finally, the ticket is closed.

Throughout the project, I will focus on configuring the ticketing system and ticket workflow. By the end of this project, I will have gained hands-on experience in IT help desk operations, improved my troubleshooting skills, and enhanced my ability to document and resolve user issues effectively.

In conclusion, this project will strengthen my technical and problem-solving skills and help prepare me to work more efficiently in an IT Help Desk role.

Network Diagram

The Ticketing system project network is designed as shown below. Windows 10 client users access the osTicket web application through a browser. When entering the osTicket domain in the browser, the request is resolved to the osTicket application hosted on the server. The osTicket application relies on three backend components, Apache, MySQL, and PHP, to function. These Services together form the backend stack. All components in the network communicate bidirectionally, allowing users to submit tickets, view updates, and interact with the system in real time.

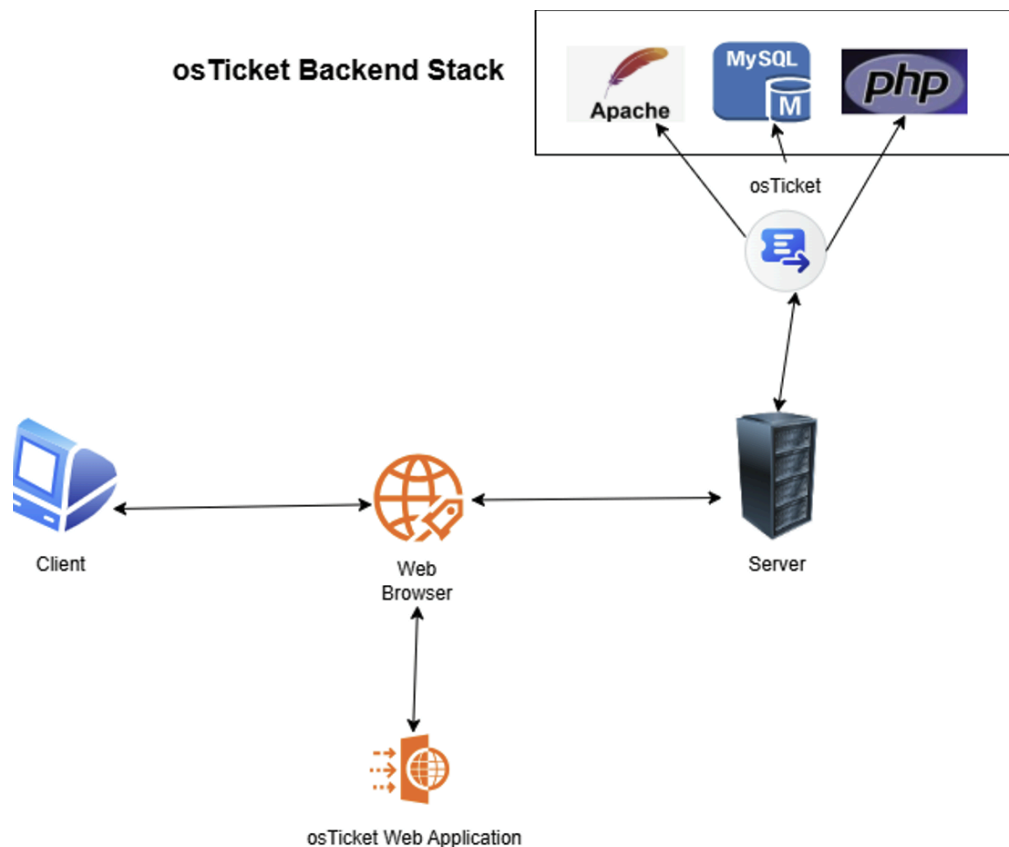


Figure 1: osTicket Network Diagram

Shows how all the components in the network communicate with each other

Workflow Diagram

The workflow of the ticketing system is straightforward.

When a user submits a ticket through the browser, the osTicket system assigns it to the IT department and automatically sends a notification to all agents.

Initially, I planned for the system to assign tickets directly to specific agents. However, this approach could lead to issues if an agent is off work or if tickets are not evenly distributed, causing some agents to become overworked.

By assigning tickets to the IT department first, the workload is balanced more effectively, and available agents can retrieve and manage tickets as needed. Once the issue is resolved, the agent updates the ticket and marks it as closed.

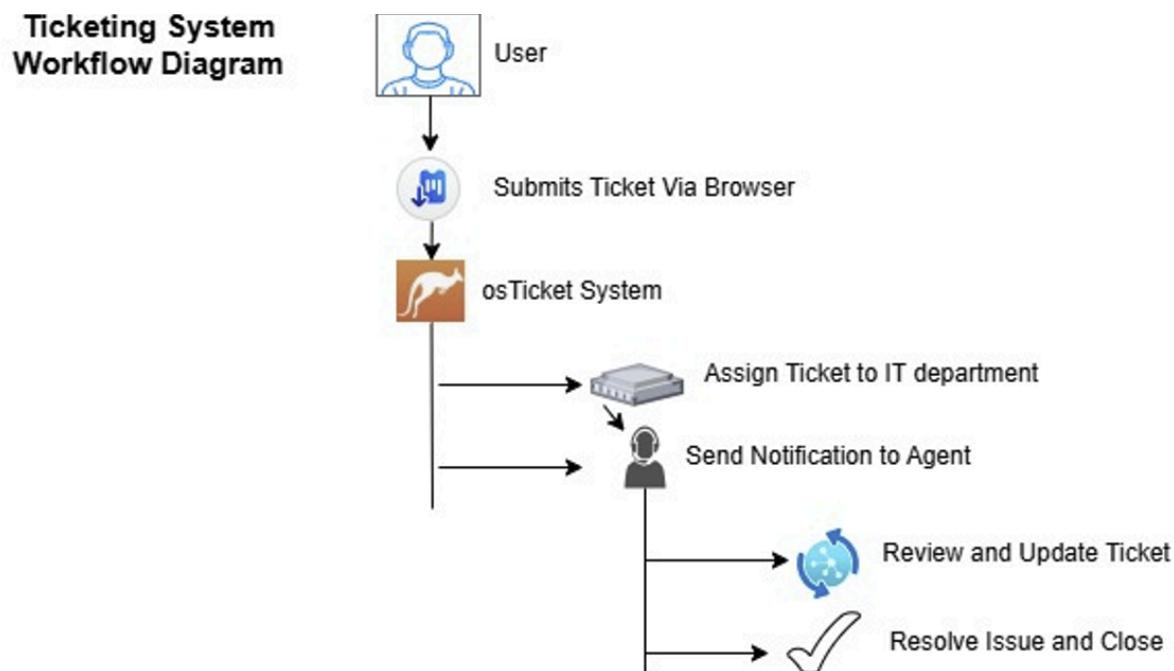


Figure 2: Ticketing System Workflow Diagram

Illustrates the stages a support ticket goes through from creation to resolution

Implementation step

Installing and Configuring XAMPP and osTicket

I Installed osTicket version 1.18 and XAMPP version 8.2.
After downloading osTicket, I extracted the folder and copied the contents of the "Upload" directory into a new folder located at C:\xampp\htdocs\osticket.

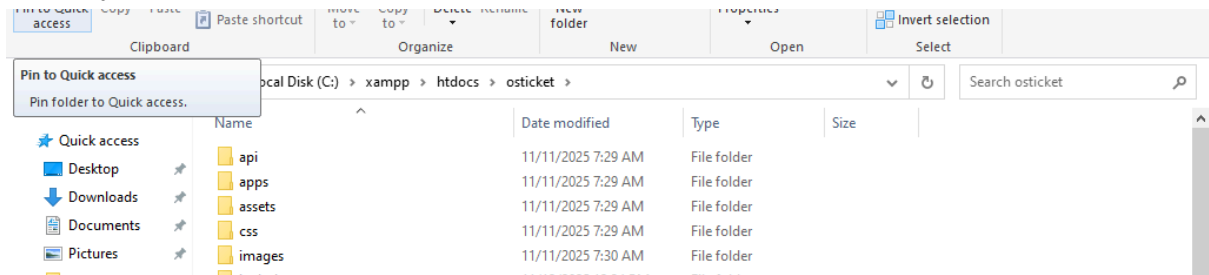


Figure 3: Contents of the upload directory moved into a new folder named osTicket inside the htdocs directory

The htdocs folder acts as the document root, meaning this is where Apache looks for files to serve in the browser.

Next, I opened the XAMPP Control panel and started both Apache and MySQL. Then, I opened a browser and typed <http://localhost/osticket>.

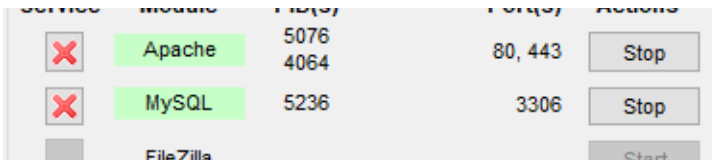


Figure 4: Both Apache and MySQL services on the XAMPP control panel

Localhost refers to the local web server running on my machine, Apache uses it to access and display files stored in the htdocs directory.

Inside the osTicket include folder, I renamed the file ost-sample-config.php to ost-config.php, as instructed by the osTicket documentation. After that, I clicked Continue to proceed with the installation wizard.

File Name	Modified	Type	Size
ost-config.php	11/4/2025 11:54 AM	PHP File	7 KB
PasswordHash	11/4/2025 11:53 AM	PHP File	7 KB
Spyc	11/4/2025 11:53 AM	PHP File	32 KB
tnef_decoder	11/4/2025 11:54 AM	PHP File	20 KB
UniversalClassLoader	11/4/2025 11:53 AM	PHP File	9 KB

Figure 5: osTicket configuration file renamed from ost-sampleconfig.php to ost-config.php

Lastly, to complete the installation, I configured the System Settings, Database Settings and Admin Account. For the system settings, I included a placeholder email address: support@localhost.local.

In the database settings, I entered my real email and specified the database I had created earlier in phpMyAdmin. Finally, I created an admin account named TheNazSec.



Installation Guide —

Congratulations!

Your osTicket installation has been completed successfully. Your next step is to fully configure your new support ticket system for use, but before you get to it please take a minute to cleanup.

Figure 6: osTicket installation successfully completed after creating the admin account and configuring the system and database settings.

Configuring System Settings

Naming the Ticket System and Creating Departments

I first started by naming my help desk ticketing system "Ticket System project." This helps identify the system, especially in an environment with multiple tools.

Steps Taken:

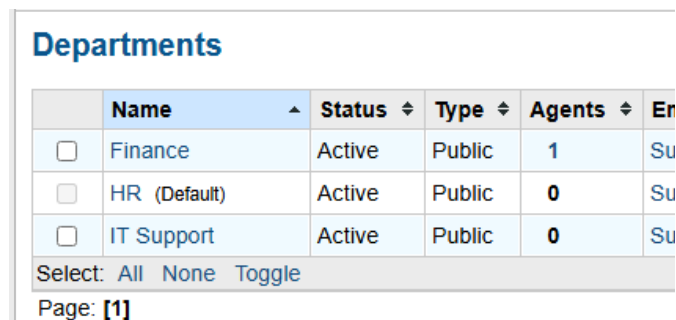
Agent Panel -- Settings -- System -- Help Desk Name

Next, I created three departments: IT Support, HR, and Finance.

In a busy organisation, hundreds of tickets come in, so separating departments helps organise and route tickets to the correct team.

Steps taken:

Admin panel - Agents - Departments - Add New Department



	Name ▲	Status ⇅	Type ⇅	Agents ⇅	Er
☐	Finance	Active	Public	1	Su
☐	HR (Default)	Active	Public	0	Su
☐	IT Support	Active	Public	0	Su

Select: All None Toggle

Page: [1]

Figure 7: Three departments created

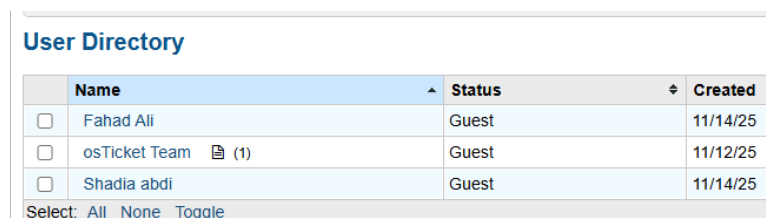
Creating User Accounts

After that, I created new user accounts.

These users are the people who will submit tickets. osTicket allows users to submit tickets without logging in, they only need to enter their name and email address.

Steps taken:

Admin Panel - Users - Add Users



	Name ▲	Status ⇅	Created
☐	Fahad Ali	Guest	11/14/25
☐	osTicket Team (1)	Guest	11/12/25
☐	Shadia abdi	Guest	11/14/25

Select: All None Toggle

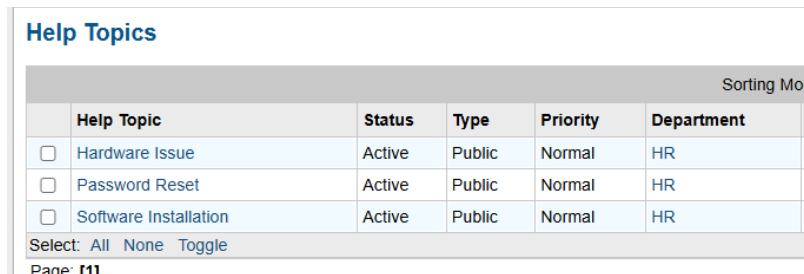
Figure 8: User accounts created

Creating Help Topics

Next, I created three Help Topics: Software Installation, Password Reset and Hardware Issues. These help topics act as the first level of sorting when a user submits a ticket. They ensure tickets go to the correct department or team and help keep the workflow organised.

Steps Taken:

Agent Panel - Manage - Add New Help topic



	Help Topic	Status	Type	Priority	Department
☐	Hardware Issue	Active	Public	Normal	HR
☐	Password Reset	Active	Public	Normal	HR
☐	Software Installation	Active	Public	Normal	HR

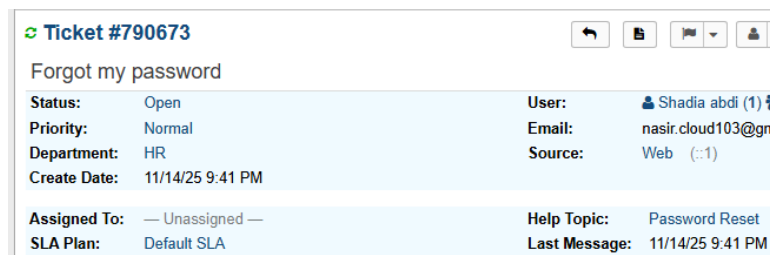
Select: All None Toggle

Page: 1/1

Figure 9: Help Topics created

Testing Workflow

Now that the new accounts, departments and help topics were created, it was time to test the ticket workflow. To do this, I submitted a test ticket as a user, selecting the appropriate help topic. After submitting the ticket, I logged back in as an agent and opened the ticket to confirm it appeared in the correct queue.



Ticket #790673	
Forgot my password	
Status:	Open
Priority:	Normal
Department:	HR
Create Date:	11/14/25 9:41 PM
Assigned To:	— Unassigned —
SLA Plan:	Default SLA
User:	Shadia abdi (1)
Email:	nasir.cloud103@gn
Source:	Web (::1)
Help Topic:	Password Reset
Last Message:	11/14/25 9:41 PM

Figure 10: Ticket successfully received after user submission

Client Machine Testing

Next, I tested the connection from the client machine. The connection didn't work immediately because of a networking issue, which I have documented in the Problems Encountered & Issue section.

From the client device, I needed to enter the server's IP address:

<http://10.0.2.10/osticket>

This IP address belongs to the server hosting Apache and osTicket.

Steps Taken:

Client Device - Browser - Entered 10.0.2.10/osticket

Once the client-side connection worked, I added two more agents, one for each department. This helps distribute the workload and prevents a single agent from being overwhelmed. The difference between agents and users is simple:

- Users submit the tickets
- Agents handle, respond to, and resolve tickets

Steps Taken:

Admin Panel - Agents - add New Agent

Click to go back (Alt+Left arrow), hold to see history

directory (../setup/) for security reasons.

Welcome, **Nasir.** | [Agent Panel](#) | [Profile](#) | [Log Out](#)

Dashboard Settings Manage Emails **Agents**

Agents Teams Roles Departments

— All Departments — — All Teams — Apply

Agents Add New Agent More

	Name	Username	Status	Department	Created	Last Login
☐	Abdi Kus	a.kus	Active	IT Support	11/15/25	never
☐	Nasir Ali	thenazsec	Active	HR	11/12/25	2 minutes ago
☐	Sandra Lucy	s.lucy	Active	Finance	11/15/25	never

Select: All None Toggle

Figure 11: Client device accessing osTicket and new agents created.

Created Service Level Agreement Plan

Lastly, I created a Service Level Agreement (SLA) plan. This helps prioritise tickets when the help desk is busy. Users can select a priority level when submitting a ticket. I created three priority levels:

- High Priority - 1 Hour Response Time
- Medium priority - 4 hour response time
- low priority - 24 hours response time

Steps taken:

Admin Panel - Manage - SLA - Add New SLA plan

Service Level Agreements

Add New SLA Plan

More

	Name	Status	Grace Period (hrs)	Date Added	Last Updated
☐	Default SLA <i>(Default)</i>	Disabled	18	11/12/25	11/15/25 7:05 PM
☐	High Priority	Active	1	11/15/25	11/15/25 7:04 PM
☐	Low priority	Active	24	11/15/25	11/15/25 7:06 PM
☐	Medium Priority	Active	4	11/15/25	11/15/25 7:05 PM

Select: All None Toggle

Figure 12: Service Level Agreement (SLA) created with three priority levels.

Workflow Testing

I created three departments in osTicket, Network Support, Hardware Support, and IT Support, with one agent assigned to each department.

To test the basic workflow, I submitted a ticket as a user from the osTicket “Open a New Ticket” page, where you enter an email address, full name, and select a Help Topic.

Steps taken:

1. Navigated to localhost/osticket → Open a New Ticket
2. Selected a Help Topic
3. Created the ticket

Next, I logged in as an agent. The new ticket appeared in the Tickets section. I opened it, added a response explaining the resolution, and then closed the ticket. The ticket status correctly changed to Closed.

This demonstrates a basic end-to-end workflow.

My plan is to simulate a real help desk environment by generating and resolving tickets every day for one month to gain realistic hands-on ticketing experience.

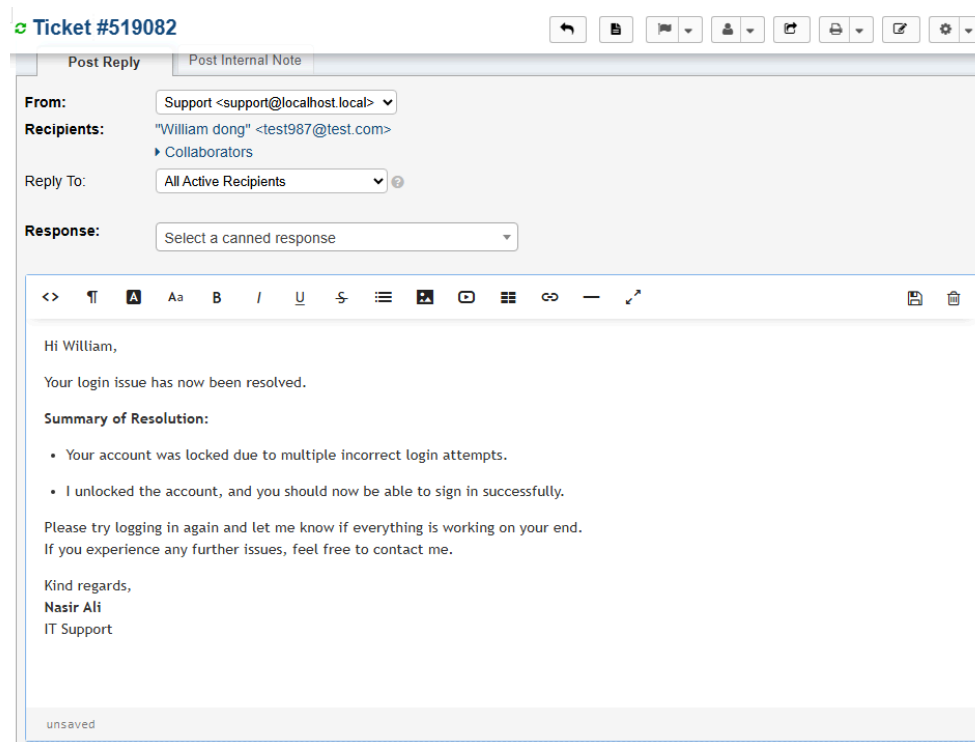


Figure 13: The agent received and resolved the ticket, then proceeded to close it.

Active Directory (LDAP) integration

Towards the end of the project, I decided to add a feature that was not part of the original plan or diagram: Active Directory LDAP authentication.

Originally, all osTicket agents logged in using accounts created manually through the osTicket admin panel. However, in a real office environment, help desk agents typically authenticate using their Active Directory Domain Services (AD DS) credentials. To make my environment more realistic and aligned with enterprise setups, I implemented LDAP integration.

osTicket cannot authenticate directly with Active Directory without a plugin. To enable LDAP support, I downloaded and installed the LDAP Authentication and Lookup plugin.

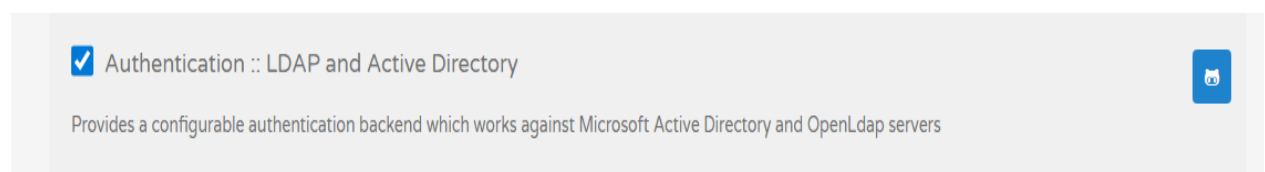
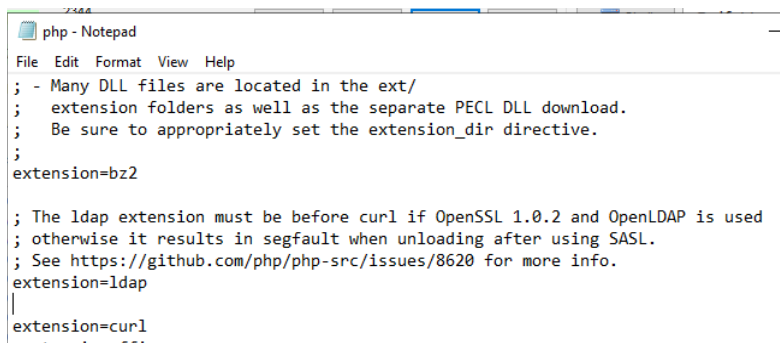


Figure 14: Downloaded LDAP and Active Directory plugin from the official osTicket website

Steps Taken

1. Visited the osTicket download page.
2. Selected “LDAP and Active Directory” and downloaded the plugin.
3. Placed the plugin folder inside /htdocs/include/plugins.
4. Edited **php.ini** and enabled LDAP by removing the semicolon (;) from:
extension=ldap



```
File Edit Format View Help
; - Many DLL files are located in the ext/
; extension folders as well as the separate PECL DLL download.
; Be sure to appropriately set the extension_dir directive.
;
extension=bz2

; The ldap extension must be before curl if OpenSSL 1.0.2 and OpenLDAP is used
; otherwise it results in segfault when unloading after using SASL.
; See https://github.com/php/php-src/issues/8620 for more info.
extension=ldap
|
extension=curl
...
```

Figure 15: Removed the semicolon from extension=ldap to enable LDAP module.

5. Opened Agent Panel - Manage - Plugins in osTicket.
6. Installed and enabled the LDAP Authentication and Lookup Plugin.
7. Configured the plugin with my Domain Controller hostname, Base DN, Bind DN, and service account credentials.

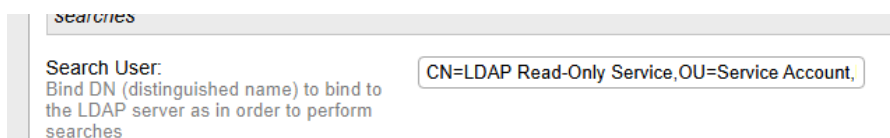


Figure 16: Before configuring the plugin, I created an user called LDAP Read-Only Service user and an organisational unit named Service Account in Active Directory (AD).

8. Tested AD DS logins and confirmed that agents can authenticate successfully.

Active Directory Preparation

Before completing the osTicket configuration, I created a new user and an organisational unit in Active Directory.

This account, named LDAP Read-Only Service, is used by osTicket to look up and read user information.

Once the user was created, I proceeded to complete the plugin configuration.

Plugin Configuration

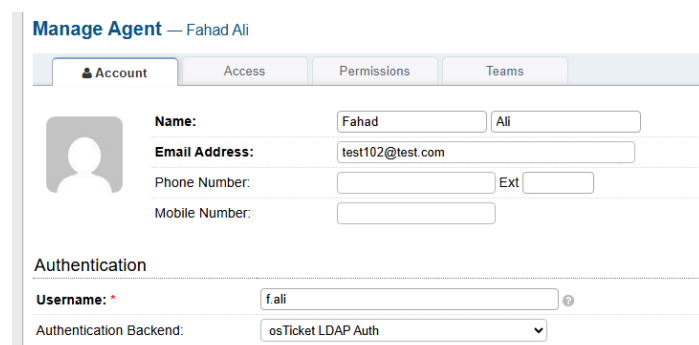
To allow osTicket to authenticate users using Active Directory, the configuration section must be filled in accurately:

- **Default Domain:** lab.local
- **LDAP Server:** 10.0.2.10
- **Bind DN:** CN=LDAP Read-Only Service, OU=Service Accounts, DC=lab, DC=local
- **Search Base:** DC=lab,DC=local

The Bind DN and Search Base tell osTicket where to search inside Active Directory for account information.

Testing Authentication

Finally, I opened an existing agent profile in osTicket and enabled LDAP authentication for that account.



The screenshot shows the 'Manage Agent' interface for a user named 'Fahad Ali'. The 'Account' tab is selected, displaying a form with the following fields: Name (Fahad Ali), Email Address (test102@test.com), Phone Number, and Mobile Number. Below the contact information is the 'Authentication' section, which includes a 'Username' field set to 'f.ali' and an 'Authentication Backend' dropdown menu set to 'osTicket LDAP Auth'.

Figure 17: A new agent was created and linked to their Active Directory account.

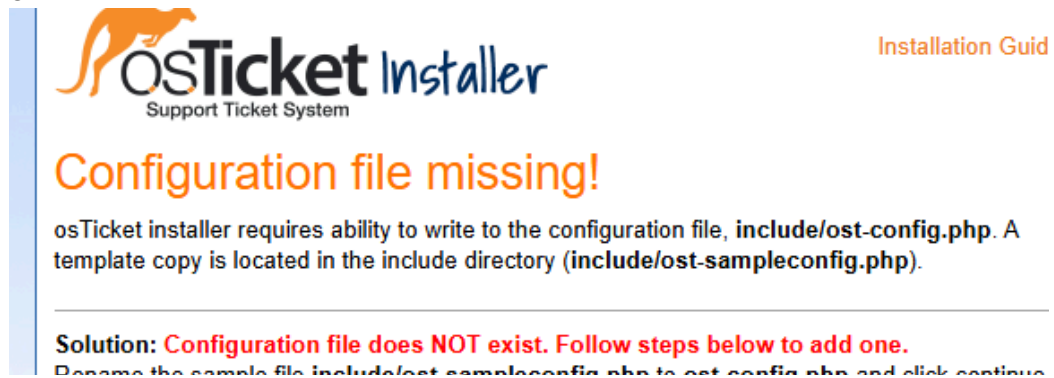
After updating the record, I attempted to log in using their Active Directory credentials.

The test was successful — confirming that osTicket can now authenticate agents directly through AD DS.

Problems Encountered & Solutions

Problem1: Configuration file does not exist

During the installation of osTicket, I received an error message on the website stating:
" Configuration file does not exist"



Steps I Took That Didn't Work:

- 1) Renamed the file from **ost-sampleconfig.php** to **ost-config.php**
- 2) Gave Full Control permission to Everyone for the file **ost-config.php**
- 3) Verified that the file was located in the correct path:
C:\xampp\htdocs\osticket\include

Discovery

In File Explorer, I enabled View - File name extensions and discovered that the file was actually named:

ost-config.php.php

This occurred because file extensions were hidden by default, and Windows automatically added an extra ".php" when renamed the file.

Resolution

I removed the extra ".php" extension, leaving the file correctly named:

ost-config.php

After saving the changes, osTicket successfully detected the configuration file and the installation proceeded without errors.

Problem 2: Client Machine unable to Access

<http://localhost/osticket>

Root Cause

localhost on the client machine refers to the client itself, not the server.

To access osTicket on the server, the client must use the server's IP address, for example:

<http://10.0.2.10/osticket>

Since port 80 traffic was blocked by the server's firewall, the client could not reach the Apache web server.

Potential Issue

1. Inbound connection from client to server is blocked.
2. Network profile or firewall rule is preventing HTTP (port 80) traffic.
3. Web server (Apache) is listening on port 80, but the firewall is not allowing traffic in.

Resolution

The issue was fixed by creating an inbound firewall rule on the server to allow HTTP traffic on port 80.